

When the Circuit Isn't There: Establishing Absence in a Behavioral Sequence Model

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Motion Sequencing tokenizes mouse behavior into a fixed vocabulary of discrete movement primitives, and prior work reports that appending an empirical first-order transition matrix to token-frequency features yields no statistically meaningful gain [1]. This result is often read as suggestive but rarely interrogated against what a sequence model computes.

Using the public moseq-drugs deposit (501 open-field sessions, 15 compounds across 6 clinical classes plus vehicle, 45 drug-dose conditions) with the tokenizer held fixed, we replicate that null and show it is stronger than reported. Scoring every model on mice absent from its training set, the transition matrix does not merely fail to help: it degrades macro-F1 from 0.667 ± 0.040 to 0.569 ± 0.048 .

Four independent manipulations then converge on the absence of any sequential signal. Hashed 2/3/4-gram features (-0.003) and a single-layer 64-unit GRU with full access to order and duration (-0.0002) are indistinguishable from frequency alone. A surrogate decomposition separates timing from ordering: destroying drug-time alignment costs -0.0288 ± 0.0070 , whereas destroying sequential structure beyond that costs $+0.0005 \pm 0.0142$. Chunk-shuffle performance is flat across block sizes 1–32. Regressing out token frequency collapses performance to 0.094 against a 0.079 floor; label permutation gives $p = 0.000$.

Which actions occur, and when they fall in the pharmacokinetic window, is a sufficient statistic for drug class in this representation.

A performance null cannot separate absent signal from inadequate features, so we ask instead whether the trained model represents order at all. It does not. Activation patching recovers no subspace whose intervention shifts the class logit beyond its random-subspace null (best normalized recovery 0.11 ± 0.06 across $k = 1-8$; null median 0.09), and Distributed Alignment Search trained to align a hidden subspace with an order-dependent variable reaches interchange-intervention accuracy 0.52 ± 0.04 against 0.50 chance.

The same pipeline finding nothing is not a method failure. On synthetic sequences from a known HMM whose two conditions have identical token frequencies by construction (Cohen's $d = 0.022$; a frequency-only model at 0.458) and differ only by an inserted duration dependency, it recovers the planted computation at interchange-intervention accuracy 0.966.

The control also yields a methodological caution: the random-subspace null is bimodal, with 8% of random 3-d subspaces matching the recovered subspace. High IIA against a mean-valued random baseline can misreport localization when that baseline is not unimodal.

Ablation without a calibrated control is a failed search effort. With one, it can serve as a measurement.

Additional Key Words and Phrases: Mechanistic Interpretability, Pose Segmentation

Conference and Archival Information

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Accepted-work listing: https://nemiconf.github.io/summer26/accepted_work.html.

References

- [1] Alexander B. Wiltschko, Tatsuya Tsukahara, Ahmed Zeine, Ricardo Anyoha, William F. Gillis, Jeffrey E. Markowitz, Ralph E. Peterson, J. Katon, Matthew J. Johnson, and Sandeep R. Datta. 2020. Revealing the structure of pharmacobehavioral space through motion sequencing. *Nature Neuroscience* 23, 11 (2020), 1433–1443. doi:[10.1038/s41593-020-00706-3](https://doi.org/10.1038/s41593-020-00706-3).